Robot data collection: operator controls

Two arms · Meta Quest 3 · operator stands behind the arms

HEADSET CONTROLLERS

Left controller → left arm

`Grip` hold to drive the left arm; let go and it stays where it is

`Trigger` closes the left gripper (release to open)

`X` hold, both grips released: **both arms go home**. Keep holding until the headset shows "at home", then release.

Right controller → right arm

`Grip` hold to drive the right arm; let go and it stays where it is

`Trigger` closes the right gripper (release to open)

`A` start recording an episode · `A` again to stop (both grips released)

Right after stopping: `A` = success, `B` = failure (6 s to answer)

`B` at other times: next camera view in the headset

To reach further: let go of the grip, bring your hand back, grip again (the arm follows hand movement, not hand position).

STAND BEHIND THE ARMS BEFORE YOU PRESS ENTER PASSTHROUGH

The direction you face at that moment becomes "forward" for the whole session. Stand behind the arms and look straight ahead, the same way the arms point. If you enter while standing beside the arms (for example facing them at 90°), every move is turned: pushing your hand forward moves the arm sideways. **If moves feel turned:** release both grips, Exit VR, stand behind the arms, Enter Passthrough again.

START OF THE DAY

1. Arms resting, nothing in reach: switch the arms on.
2. Open the app: **Robot data collection** icon (or `localhost:8440`).
3. Plug the Quest in by USB. Collect → Headset → **Open page in headset** (or open `localhost:8443` in the Quest browser).
4. Bench → Checks → **Read-only arm check** (no motion; valid 8 h).
5. Collect → Recording: choose the **task** and type your **name**.

END OF THE DAY

1. **Stop arms**; switch the arms off once they are at rest.
2. Episodes → **Run checks** on the day's session.
3. Accept or reject each episode (look at the preview and the verdict).
4. Jobs → name the dataset → **Start export**.

Never run checks or exports while recording or while the arms run.

EACH RUN

1. Collect → Arms → **Live** → **Start arms**; tick the checklist (arms at rest, someone at the power switch).
2. Headset page: Input mode **Bimanual** (set when opened from the app). **Behind the arms**, press **Enter Passthrough**. Release both grips: the arms go home.
3. `A` record → do the task → `A` stop → `A`/`B`.
4. Between episodes: hold `X` to bring both arms home.
5. **Stop arms** when done. The arms return to rest. A run also ends by itself at its time limit.

WHEN THE ARMS HOLD STILL

Grip released	Normal: that arm waits. Grip again to continue.
Arms too close	Move the hands apart; the arms continue.
Gripper at the table	Lift the hand slightly; the arm continues.
Tracking lost	Look at your hands so the headset sees both controllers.

SAFETY

- **The arm power switch is the emergency stop.** Someone stands at it whenever the arms run.
- **Stop arms** (red, top of the app) brings both arms back to rest. Press it once and wait.
- `SAFETY STOP`: the program stopped the run and returns the arms to rest by itself. Wait until they are at rest, then start a new run.
- `CUT ARM POWER NOW`: switch the arms off at the power switch, then tell the person in charge.
- Keep hands, faces and objects out of the arms' reach while they move. Never clear arm errors yourself.